Testing Part 2: 6 channel test

Objective: test the functionality of all 6 channels

Setup: replicate the single channel test below and expand to test all 6 channels

Schematic:

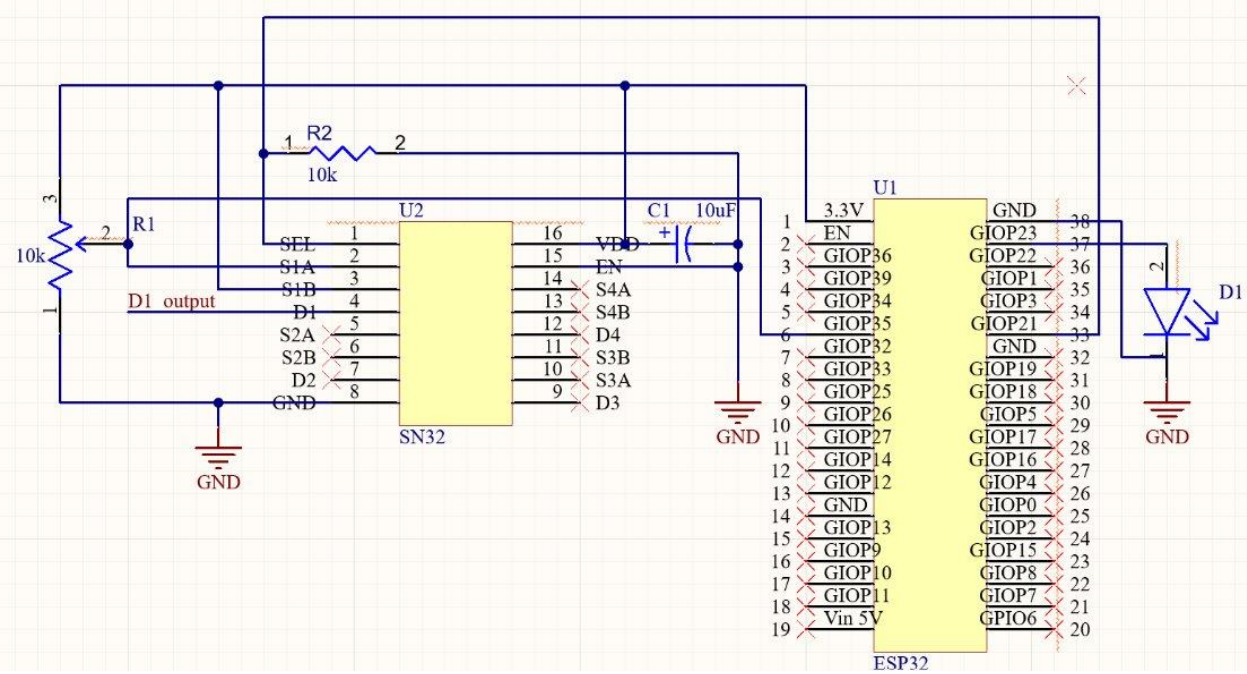


figure 1. the schematic used for single channel test

- Measure the voltage reading at SEL, S1A, S1B and D1. Repeat these measurements for all pins being used, D1-D6
- Record the potentiometer position

TEST 3:

Objective: implement current takeover logic to the SN32 with all 6 channels.

Setup:

Incorporate ESP decision-making on the SEL pin. Use PS5 to make inputs to S1B, then make manual inputs to S1A and test to see if the ESP can detect the change in inputs and route the manual signal.

Schematic

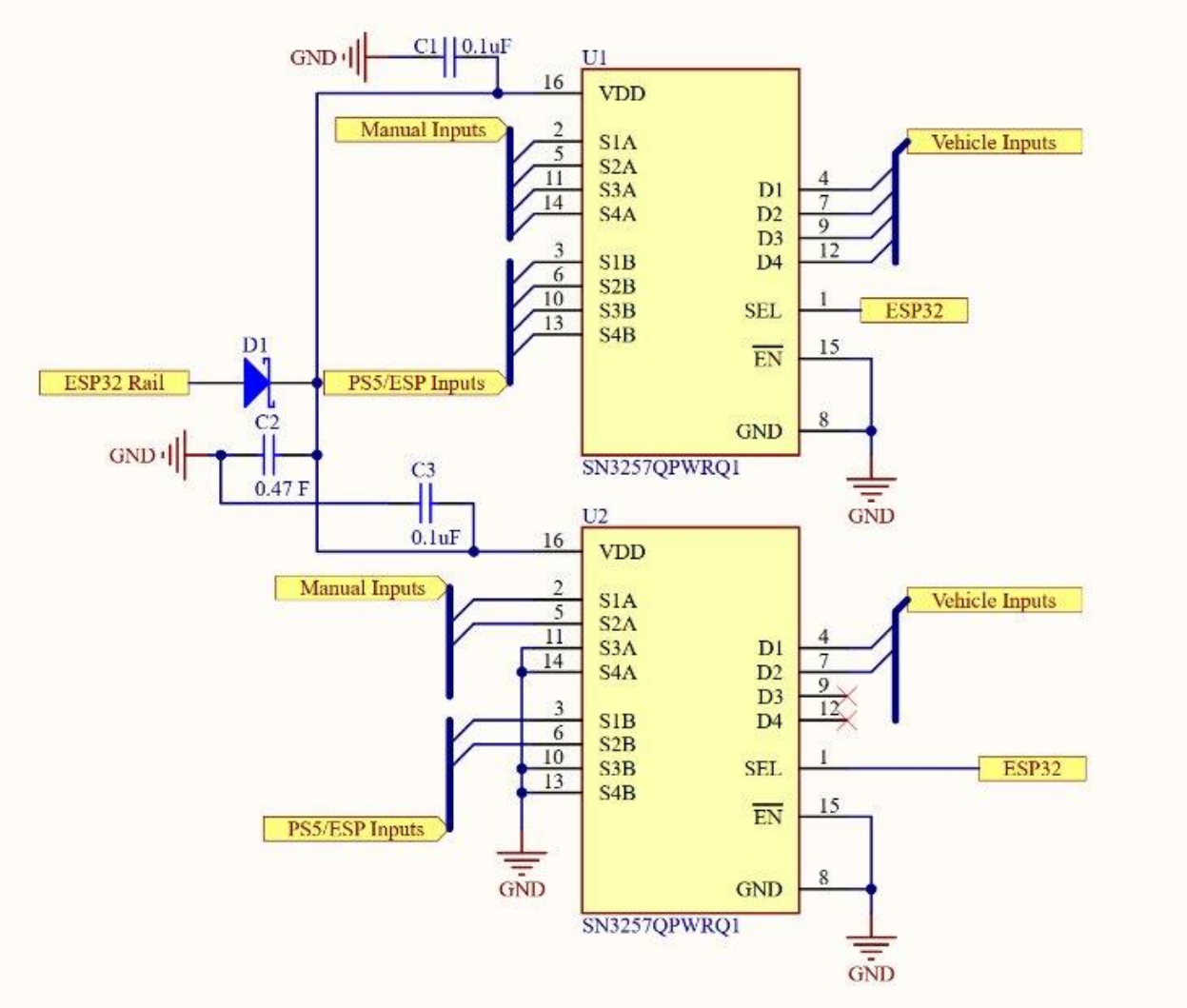
Successful completion:

- Able to demonstrate the ESP changing the SEL state using the current takeover logic

TEST 4: supercapacitor

Objective: simulate power loss of the ESP32 and/or the SN32 and observe how the analog switch behaves when board losses power and if the supercapacitor can supply enough power for long enough

Schematic:



Successful completion: We have an exact amount of time we have after power loss to operate manually.